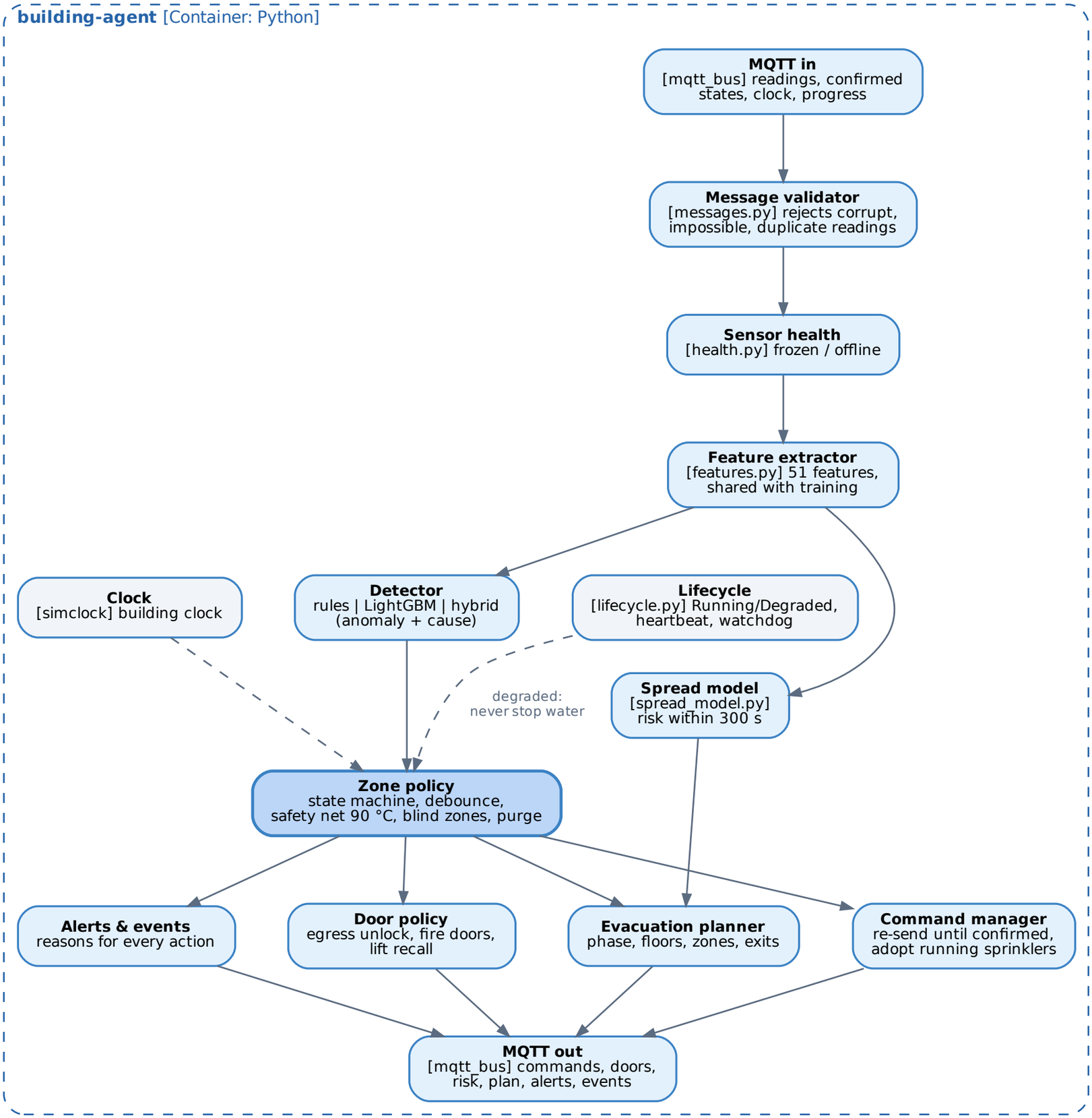
building-agent [Container: Python]
MQTT in
[mqtt_bus] readings, confirmed states, clock, progress
Message validator
[messages.py] rejects corrupt, impossible, duplicate readings
Sensor health
[health.py] frozen / offline
Feature extractor
[features.py] 51 features, shared with training
Clock
[simclock] building clock
Detector
rules | LightGBM | hybrid (anomaly + cause)
Lifecycle
[lifecycle.py] Running/Degraded, heartbeat, watchdog
degraded: never stop water
Spread model
[spread_model.py] risk within 300 s
Zone policy
state machine, debounce, safety net 90 °C, blind zones, purge
Alerts & events
reasons for every action
Door policy
egress unlock, fire doors, lift recall
Evacuation planner
phase, floors, zones, exits
Command manager
re-send until confirmed, adopt running sprinklers
MQTT out
[mqtt_bus] commands, doors, risk, plan, alerts, events